

Epidemiological trends strongly suggest exposures as etiologic agents in the pathogenesis of sporadic Alzheimer's disease, diabetes mellitus, and no...

Nitrosamines mediate their mutagenic effects by causing DNA damage, oxidative stress, lipid peroxidation, and pro-inflammatory cytokine activation, which lead to increased cellular degeneration and death. However, the very same pathophysiological processes comprise the "unbuilding" blocks of aging and insulin-resistance diseases including, neurodegeneration, diabetes mellitus (DM), and non-alcoholic steatohepatitis (NASH). Previous studies demonstrated that experimental exposure to streptozotocin, a nitrosamine-related compound, causes NASH, and diabetes mellitus Types 1, 2 and 3 (Alzheimer (AD)-type neurodegeneration). Herein, we review evidence that the upwardly spiraling trends in mortality rates due to DM, AD, and Parkinson's disease typify exposure rather than genetic-based disease models, and parallel the progressive increases in human exposure to nitrates, nitrites, and nitrosamines via processed/preserved foods. We propose that such chronic exposures have critical roles in the pathogenesis of our insulin resistance disease pandemic. Potential solutions include: 1) eliminating the use of nitrites in food; 2) reducing nitrate levels in fertilizer and water used to irrigate crops; and 3) employing safe and effective measures to detoxify food and water prior to human consumption. Future research efforts should focus on refining our ability to detect and monitor human exposures to nitrosamines and assess early evidence of nitrosamine-mediated tissue injury and insulin resistance.